

PLYMOUTH MOUNT BATTEN, United Kingdom

WMO: 038270

Lat: 50.3550N

Lon: 4.1211W

Elev: 27

StdP: 101.00

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.2	0.0	-5.6	2.4	0.9	-3.9	2.7	1.7	19.2	9.6	17.2	8.8	3.3	70	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.5	23.5	17.9	21.8	17.1	20.4	16.5	18.7	21.9	17.9	20.6	17.2	19.4	4.7	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	12.6	19.6	16.9	12.1	18.6	16.4	11.7	18.1	53.0	22.1	50.6	20.8	48.6	19.5	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.1	13.2	11.5	-3.3	26.4	1.5	1.9	-4.3	27.8	-5.2	29.0	-6.0	30.0	-7.1	31.5
			-3.9	20.6	1.6	1.6	-5.1	21.7	-6.0	22.7	-6.9	23.6	-8.0	24.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	6.6	6.5	7.6	9.4	12.0	14.6	16.5	16.6	14.9	12.4	9.2	7.3
	DBStd	4.38	2.84	2.58	2.36	2.36	2.35	2.11	1.96	1.85	1.83	2.30	2.71	3.02
	HDD10.0	468	106	98	78	36	7	0	0	0	7	46	91	
	HDD18.3	2633	363	331	332	267	197	113	64	61	104	184	274	343
	CDD10.0	896	2	1	4	19	68	139	202	204	146	81	22	7
	CDD18.3	17	0	0	0	0	0	3	7	6	1	0	0	0
	CDH23.3	64	0	0	0	0	1	16	26	20	1	0	0	0
	CDH26.7	5	0	0	0	0	0	1	1	3	0	0	0	0
(14) Wind	WSAvg	5.2	6.3	5.8	5.5	4.9	4.9	4.7	4.5	4.4	4.6	5.6	5.5	6.1
(15) Precipitation	PrecAvg	1004	112	91	85	61	62	56	56	69	80	99	110	119
	PrecMax	1333	215	205	178	153	125	145	133	141	182	210	227	229
	PrecMin	740	13	2	3	4	4	5	8	6	13	7	40	24
	PrecStd	148	56	53	40	34	35	38	30	38	46	49	49	54
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	12.3	15.4	19.6	22.0	25.7	26.0	26.0	22.5	19.0	15.0	13.2
		MCWB	11.2	10.2	11.0	13.9	16.6	18.1	18.8	19.4	17.3	16.0	13.8	12.4
	2%	DB	11.5	11.2	13.1	16.3	19.4	22.1	23.6	22.7	20.5	17.1	14.1	12.6
		MCWB	10.8	10.0	10.0	12.0	14.8	16.5	17.8	18.0	16.6	14.7	13.2	11.9
	5%	DB	11.0	10.6	11.7	14.3	17.6	20.0	21.9	21.0	19.1	16.2	13.6	12.1
		MCWB	10.3	9.6	9.5	10.8	13.9	15.6	17.3	17.2	15.8	14.4	12.7	11.4
	10%	DB	10.5	10.1	10.8	13.0	16.0	18.5	20.3	19.7	18.0	15.5	13.1	11.5
		MCWB	9.7	9.1	9.2	10.1	12.9	14.9	16.6	16.6	15.1	14.1	12.2	10.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	11.2	11.7	14.4	17.2	19.0	19.6	20.7	18.4	16.6	14.4	12.7
		MCDB	11.8	11.8	13.7	18.7	20.5	24.1	24.0	23.7	20.8	18.0	14.6	12.9
	2%	WB	11.1	10.6	10.9	12.4	15.5	17.2	18.6	18.8	17.4	15.7	13.6	12.2
		MCDB	11.3	10.9	12.0	15.2	18.5	20.9	22.2	21.4	19.2	16.4	13.9	12.5
	5%	WB	10.6	10.0	10.2	11.4	14.4	16.3	17.8	18.0	16.7	15.1	13.0	11.7
		MCDB	10.9	10.4	11.1	13.4	16.8	19.2	21.0	20.0	18.1	15.7	13.5	12.0
	10%	WB	10.0	9.4	9.7	10.7	13.4	15.4	17.1	17.3	16.1	14.6	12.4	11.0
		MCDB	10.4	9.9	10.5	12.3	15.4	17.7	19.5	19.0	17.2	15.2	12.9	11.4
(35) Mean Daily Temperature Range	5% DB	MDBR	4.3	4.8	5.5	6.4	6.6	6.6	6.5	6.5	6.4	5.2	4.8	4.4
		MCDBR	3.9	4.8	6.6	8.3	8.7	9.3	9.2	8.1	7.7	5.7	4.3	3.8
	5% WB	MCWBR	4.2	4.1	4.6	5.3	5.5	5.4	5.0	4.7	4.7	4.0	4.0	4.1
		MCDWR	3.8	4.3	5.3	6.9	7.8	7.9	7.8	6.9	6.4	4.6	3.8	3.6
(40) Clear-Sky Solar Irradiance	taub	taub	0.335	0.349	0.379	0.390	0.396	0.385	0.385	0.372	0.367	0.359	0.333	0.326
		taud	2.352	2.371	2.297	2.293	2.318	2.381	2.387	2.440	2.441	2.431	2.407	2.321
	Ebn at Noon	Ebn at Noon	685	780	817	851	861	873	866	861	826	759	686	630
		Edh at Noon	67	84	107	119	121	115	113	102	92	78	63	60
(44) All-Sky Solar Radiation	RadAvg	0.81	1.59	2.77	4.44	5.37	5.88	5.46	4.63	3.50	1.97	1.04	0.65	
	RadStd	0.07	0.16	0.32	0.33	0.34	0.55	0.38	0.33	0.23	0.13	0.09	0.08	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page